

Prithwjit Ghosh

Data Scientist Specialist | Accenture Global Technology

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GitHub | Portfolio | LinkedIn

Data Scientist with 3 years of experience and an M.Sc. in Statistics from **IIT Kanpur** (AIR 7), specializing in production-grade ML systems, time-series forecasting, and agentic GenAI across enterprise platforms serving **70,000+** contracts and multi-million customer bases.

Professional Experience

Data Scientist Specialist | *Accenture Global Technology*
Bengaluru, Karnataka, India

July 2023 – Present

● Diamond Projects

- **SALES AND GUEST-COUNT FORECASTING** | *Global Quick Service Restaurant (QSR) Brand*
 - Architected and productionized an end-to-end ensemble forecasting system using Prophet, Theta, MSTL, LightGBM, and Naive models across **6** countries to accurately predict sales and guest counts up to **48 months** into the future
 - Developed a custom horizon-aware evaluation framework spanning **450+** forecasting models and a weighted ensemble balancing strategy, ultimately achieving **97-99%** accuracy at **24 months** and **95-96%** accuracy at **48 months**
 - Established a production-grade **MLOps** pipeline with **Docker** containers, GitHub Actions **CI/CD**, and SonarQube/Snyk **security** scanning, running fully automated monthly forecasts through Airflow **DAGs** orchestrated on AWS **Fargate** infrastructure
- **INTELLIGENT COLLECTIONS 3.0** | *Water Treatment Brand*
 - Trained customer-level late payment prediction models using **XGBoost**, selecting the **top 50** optimal features from **1000+** candidates and applying **quantile calibration** to reach **84-90% AUC** across a **10,000+** monthly customer base
 - Formulated short-term accounts receivable forecasting models using both statistical and ML/DL methods, consistently reaching **95-98% accuracy** and delivering results to leadership through interactive **Power BI** dashboards
 - Deployed calibrated risk scores integrated with accounts receivable forecasts into collections workflows, reducing overdue amounts by **38%**, lowering open accounts receivable by **15%**, and increasing collections by **12%** over two years
- **FINANCIAL FORECAST INTELLIGENCE** | *Accenture Technology*
 - Spearheaded development of a natural language decision intelligence platform over contract profit and loss data, answering finance managers' questions on margin and cost drivers across **70,000+** active contracts
 - Orchestrated a scalable multi-agent system built on LangChain, LangGraph, Langfuse and **Gemini 2.5 Flash** featuring a **typed intent compiler**, **8** domain specialist agents, and a **53-node** causal knowledge graph
 - Replaced traditional language model routing with deterministic **graph traversal** to enable reproducible, source-traced reasoning, dramatically cutting variance analysis time from several hours down to just **10 seconds**

● Short Projects

- **Machine Failure Alert System** | *Global QSR Brand*
 - Designed a shake machine failure alert system with a statistical and **XGBoost/Decision Tree** ensemble (**99% accuracy, 92% recall**) vs. 16% baseline, extending failure lead time from **3-5** to **30 minutes** for early intervention
 - Architected the full production pipeline end-to-end using **Kafka**, GCP, and S3 for ingestion, GCP-based training with daily inference on **edge servers**, **Prometheus/Grafana** for monitoring, and **NATS** for real-time alert delivery
- **Cash Flow Forecasting** | *Power Utility Company*
 - Delivered end-to-end **B2C** cash flow forecasting for **6M+** customers using advanced ML modeling, improving **Cash-In** accuracy from **70% to 98%** and **Cash-Out** from **64% to 93%**, with insights surfaced through interactive dashboards
- **Marketing Data Analysis** | *Internationally Renowned Liquor Brand*
 - Analyzed customer behavior across subscription patterns, campaigns, **conversion funnels**, **up-sell**, **cross-sell**, and **churn propensity** using **RFM** and advanced analytics to drive data-informed growth and retention strategies

Skills

- **Languages:** Python, SQL (MySQL, Redshift, Presto, DuckDB, SQLite), R
- **Microservices/ API:** FastAPI, Redis, Prometheus, Grafana, Kafka, NATS
- **ML/DL Libraries:** Scikit-learn, XGBoost, LightGBM, Prophet, TensorFlow, Keras, NLTK, spaCy
- **GenAI/LLM:** LangChain, LangGraph, RAG, AI Agents, Agentic AI, OpenAI, Claude, Claude Code, Ollama
- **Cloud/ Deployment:** AWS (Fargate, EMR, EC2, S3, CloudWatch), GCP, Docker, Airflow, Astronomer, GitHub Actions
- **BI/ Analytics Tools:** Power BI, Streamlit, MS Excel, MS PowerPoint
- **Domains:** Predictive Modeling, Time Series Forecasting, Classification, NLP, Statistics, A/B Testing, MLOps

Education

- **Master of Science in Statistics**, Indian Institute of Technology Kanpur (IITK) Aug '21 - Jul '23
CGPA: 8.9 | JAM Rank: 7 (AIR) | Department Rank: 6
- **Bachelor of Science in Statistics**, Bidhannagar College Jun '18 - Jul '21
CGPA: 9.99 | Department Rank: 1